# bGeigieZen Workshop Description

Safecast logo

## bGeigieZen Build Workshop — Hands-On Radiation Monitoring

**Date:** 2026-08-14 | **Version:** 1.0

A one-day workshop for 10 students combining DIY electronics with environmental science, in the spirit of Safecast's open, participatory citizen-science model.

### Age Group

Recommended for **ages 13+ (middle/high school and up)**. The build requires fine motor control for soldering and following multi-step instructions; Safecast's past workshops have run successfully from middle-school students through university and adult learners. For younger students (elementary), Safecast's simpler no-solder **kGeigie** kit is a better fit — ask us about that option.

### Learning Objectives

By the end of the workshop, students will be able to:

- **Explain core radiation concepts**: distinguish ionizing radiation, background vs. contamination, and read/interpret CPM and µSv/h units.
- **Understand why continuous, geolocated monitoring matters**, using Safecast's post-Fukushima origin as a case study in citizen science.
- **Identify and use basic electronics tools and practices**: soldering, wiring, connector/header handling, and static-safe workspace habits.
- **Assemble a working embedded device** from a kit, following a schematic/build guide, and diagnose simple assembly faults (bad joint, loose connector, GPS lock failure).
- **Collect and interpret real sensor data**: verify a calibration reading, log a GPS-tagged radiation track, and read live values off the device display.
- **Contribute data to an open dataset**: upload a field log to the Safecast map/API and describe what the resulting data shows about the local environment.

## Format (approx. 4–5 hrs)

1. **Intro to radiation (30 min)** — What radiation is, units (CPM/μSv/h), background vs. contamination, why open data matters (Fukushima origins of Safecast).
2. **Build session (2–2.5 hrs)** — Each student assembles their own bGeigieZen: GPS, Geiger tube, LCD, soldering/wiring, case assembly. Pairs help troubleshoot.
3. **Power-up & calibration (30 min)** — Boot devices, verify GPS lock, compare CPM readings against a reference source.
4. **Field walk (30–45 min)** — Log a short GPS-tagged radiation track around the venue; see live readings on the LCD status bar.
5. **Wrap-up (30 min)** — Upload logs to the Safecast API/map, discuss what the data shows, Q&A on nuclear basics and open data.

## Tools Needed

- Flush cutters and nippers (for cutting pin headers and excess wire)
- Temperature controlled soldering iron (fine tip)
- Small Phillips-head screwdriver
- Small flat-head screwdriver
- USB-C cable for programming and charging (not included in the kit)

### Workspace

- Well-lit workbench
- Silicone rubber mat (antistatic)

## What We Need From the School

- **Room**: a classroom or lab with 5+ tables (2 students/table), seating for 10, and good ventilation (soldering fumes).

- **Power**: at least 5 grounded power outlets/extension cords (2 students share a soldering iron station).

- **Internet/Wi-Fi**: needed for the wrap-up data-upload step. Or for remote Safecast support

- **Adult supervision**: one teacher/staff member present throughout, in addition to the workshop instructor, for duty-of-care and to know the students.

- **Basic safety**: fire extinguisher or fire blanket accessible (standard for any soldering activity), and eye protection if the school has it.

## Support

If budget allows, a Safecast volunteer expert can be made available to help with the build, either in person or remotely.

# Takeaways

Each student leaves with a working, GPS-logging Geiger counter, a basic understanding of ionizing radiation, and a first contribution to Safecast's open dataset.

# Pricing

Prices per unit from the bGeigieZen store:

| Item | Price |
|---|---|
| bGeigieZen Kit (for assembling) | $475 |
| bGeigieZen Assembled | $750 |
| bGeigieZen Cradle Kit (unassembled) | $60 |
| bGeigieZen Cradle (assembled) | $80 |

**Estimated total for a 10-student workshop:**

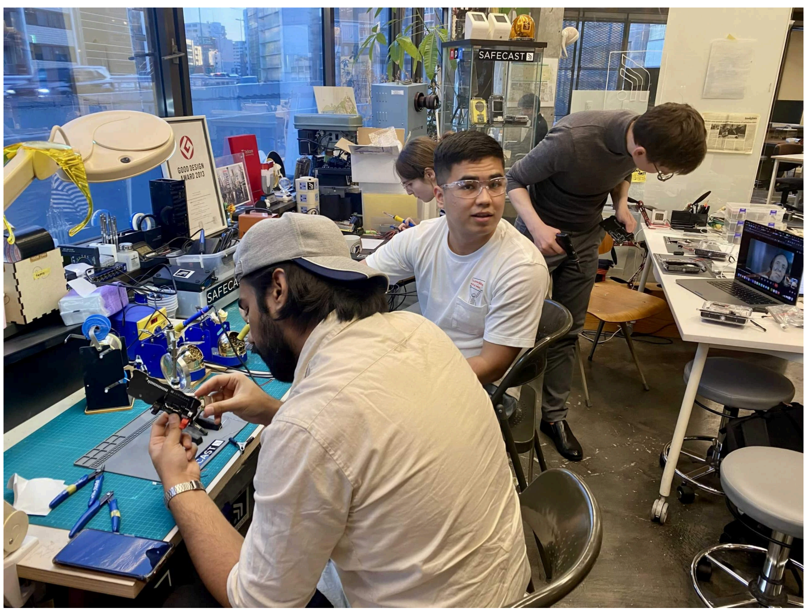
| Item | Qty | Cost |
|---|---|---|
| bGeigieZen Kit (10-unit discount, 10% off) | 10 | ~~$4,750~~ **$4,275** |
| Shipping/import | — | Varies by destination, quoted separately |
| Volunteer instructor (if not local) | 1 | Travel/accommodation only, or free if remote |

**A 10% discount applies when ordering 10 or more bGeigieZen kits together**, bringing the workshop set to **$4,275** ($427.50/kit). Optional cradle kits ($60 each) can be added if devices will be left running as fixed stations after the workshop. Prices are subject to change — check the store page or confirm with Safecast for current rates. Schools/organizers cover kit cost + shipping; Safecast does not currently charge a separate workshop fee.

# Links

- bGeigieZen product page (safecast.org)
- bGeigieZen website & store (bgeigiezen.safecast.jp)
- Safecast introduces the bGeigieZen
- Safecast 5th Anniversary bGeigie Workshop — Tokyo
- Upcoming events in Hong Kong and Taipei
- Safecast Events archive
- Minerva University Students Reimagine the Safecast Map (Tokyo)
- bGeigieZen News & Updates (bgeigiezen.safecast.jp) — Hong Kong (2024), Spain/Ibercivis (2024), Amsterdam University of Applied Sciences (2024), and Minerva University Tokyo (2026) workshops

# Workshop Photos


*Photos from Safecast's Minerva University workshop, Tokyo.*